Performance and cost experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which cases, timing boundary and quality rule the comparison covers.

Teach raw records, central summaries and resource constraints without benchmark or pricing overclaims.

Keep one habit: Model main middle pairs → inspect aggregate means → apply inclusive mean budget → assess C/D quality → record missing repeated-run evidence.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of performance and cost experiments

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of accessibility and user feedback

Reference example, not the only acceptable wording: Teacher explanation: Accessibility makes information and controls usable across stated needs.

Feedback should describe the problem and recovery action without exposing sensitive details. A form provides visible labels, keyboard access, associated error text and a non-colour status indicator.

Test at zoom and with keyboard. Do not treat automated checks as a complete user evaluation.

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of incident investigation

Reference example, not the only acceptable wording: Teacher explanation: Incident investigation preserves a timeline and distinguishes **observation** from hypothesis. Repair before **evidence** collection can erase clues.

Record timestamps, versions, requests, authorised actions and outcomes with minimised logs. Separate immediate containment from diagnosis. Compare candidate explanations against **evidence**.

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of deployment environments

Reference example, not the only acceptable wording: Teacher explanation: Development, staging and production are distinct environments with explicit data and credential boundaries.

Test configuration must not silently target production. For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names.

A staging run cannot use production write credentials.

Pictured end goal

Explain performance and cost experiments using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Accessibility and user feedback

Incident investigation

Deployment environments

Use the three-session teaching cycle above, with extra time as needed. Performance experiments compare **matched workload**s with declared timing boundaries and quality constraints. Averages can hide slow cases. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of performance and cost experiments

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach matched records, multiple summaries and mandatory constraints

> Separate the main four-case timing data from the six-case aggregate picture.

Definitions and complete supplied-observation contract

Performance experiments compare **matched workload**s with declared timing boundaries and quality constraints. Averages can hide slow cases.

Use the same cases and environment; record every duration and success. For a simple **median** sort values and average the middle pair for even counts. Keep failed requests visible.

Experiment item	Complete supplied-fixture rule
Matched main workload	Ordered fictional cases Q1–Q4; same environment and rubric assumed
Timing boundary	Supplied seconds from request start to final outcome; includes failed outcomes in added probe
Main A durations	[1, 1, 1, 9] seconds; 4/4 pass
Main B durations	[2, 2, 2, 2] seconds; 4/4 pass
Mean	Sum of all recorded durations / number of recorded cases
Median with four cases	Sort; average positions 2 and 3

Experiment item	Complete supplied-fixture rule
Tail disclosure	Show every duration and observed maximum; four cases do not establish a population tail percentile
Separate picture	Six questions each; A total 18 seconds/12 units, B 12 seconds/18 units; supplied equivalent quality
Changed picture budget	Mean cost at most 2 fictional units per case; totals cannot establish every individual request cost
Limits	Constructed observations, no actual benchmark, repeated runs, prices or universal winner

Main summary ledger and checked reasoning

Main supplied measure	A: [1, 1, 1, 9] seconds	B: [2, 2, 2, 2] seconds
Total / count	12 / 4	8 / 4
Mean seconds	3	2
Middle pair after sorting	1 and 1	2 and 2
Median seconds	1	2
Observed maximum seconds	9	2
Quality on these cases	4/4 pass	4/4 pass
Scoped comparison	Lower median; one slow 9-second case	Lower mean and observed maximum; no universal preference without objective

Main four cases: A mean 3, median 1, observed maximum 9 seconds; B mean/median/maximum 2 seconds; both 4/4 quality. Picture separate six cases: A mean 3 seconds/2 fictional cost units; B 2 seconds/3 units. Changed budget is explicitly mean cost ≤ 2, so A meets equality and B fails; no individual cap follows from totals. Added failure probe retains its failed duration and is distinct from reserved C/D.

Selected C mean 4, median 2.5; D mean/median 3 with 3/4 quality. Report D's failure and do not infer C's quality beyond what the task supplies. All data are constructed, with no actual benchmark, population-tail or live-pricing claim.

Performance and cost experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and median.	Use the same cases and environment; record every duration and success. For a simple median sort values and average the middle pair for even counts. Keep failed requests visible.	A mean 3, median 1; B mean 2, median 2. B lowers mean but A has lower median; show the 9-second tail instead of declaring one universal winner.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

Qualification: Resource improvement under one workload is not universal efficiency or equal quality.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**.

A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Quality, workload and mean-budget probes

Performance probe	Expected interpretation
Mean improves while median worsens	Report both and raw durations; no universal winner
One slow case removed from report	Biased summary; preserve all supplied eligible outcomes
Fast response is wrong	Keep its duration and report quality failure; speed is not successful task completion
Mean budget met at equality	Allowed under at-most rule
Total cost/count within budget	Establishes mean only, not an individual-request maximum
Different case IDs/order	Reject paired fixture comparison until workload is matched
Negative, NaN or Boolean duration	Reject invalid supplied measurement
Four cases, no repeated trials	No stable tail behavior or precision claim
Latency/cost preference unclear	State the objective and mandatory constraints before choosing

Separate aggregate picture and missing-distribution limits

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

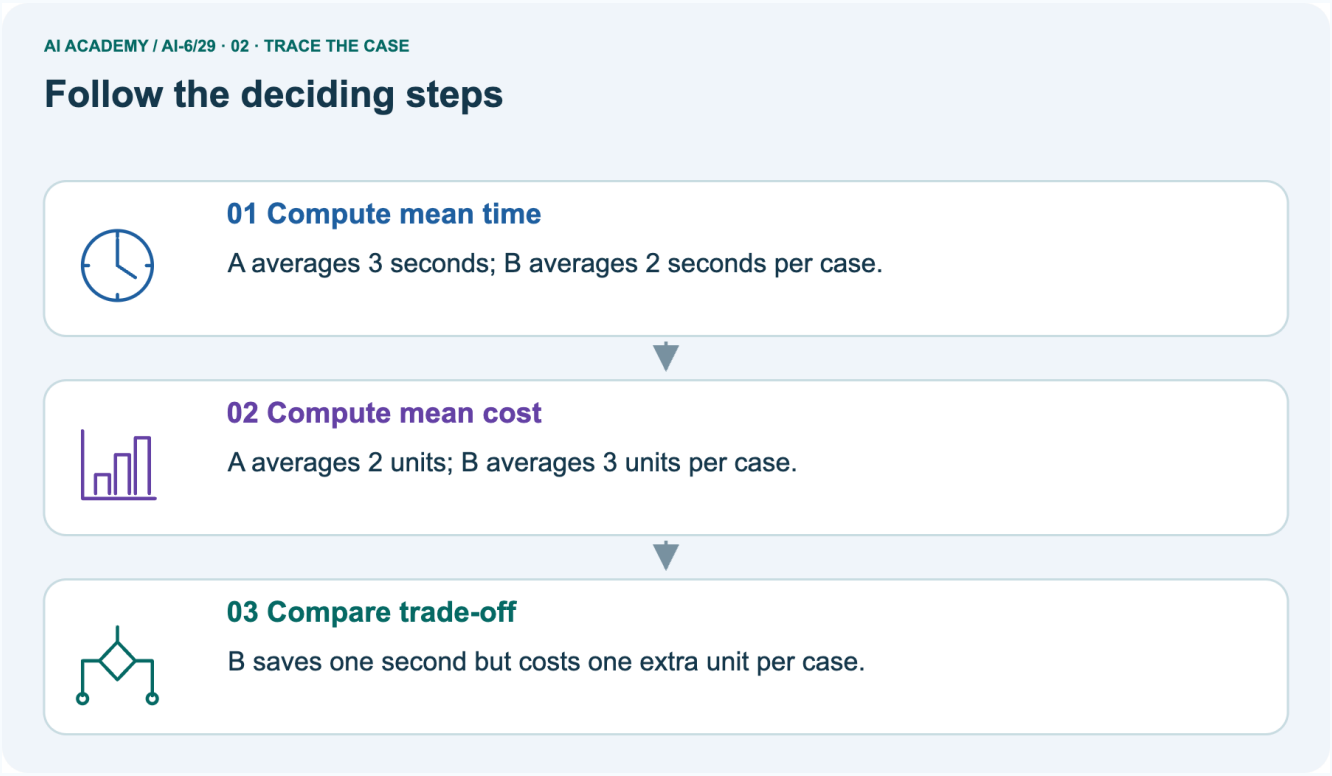
A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

The picture is a separate six-question aggregate, not a continuation of the main four durations. A averages 18/6 = 3 seconds and 12/6 = 2 cost units; B averages 12/6 = 2 seconds and 18/6 = 3 units. B saves one second per case on average while using one extra cost unit. Under a mean-cost budget of at most 2 units, A meets equality and B fails.

Individual request costs and durations are not supplied for this picture, so do not claim an individual cap, median or maximum. Cost units are fictional, not money. Quality is supplied as meeting the same rubric; broader reliability and repeated-run behavior remain unmeasured.

Separate picture: six cases each	**A**	**B**
Total seconds / cases	18 / 6	12 / 6
Mean seconds per case	3	2
Total cost units / cases	12 / 6	18 / 6
Mean cost units per case	2	3
Mean budget ≤ 2 units	Meets inclusive boundary	Fails
Not inferable from totals	Individual max, median, cost cap	Individual max, median, cost cap

Picture trace and inclusive mean-budget key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute mean time	A averages 3 seconds; B averages 2 seconds per case.
2. Compute mean cost	A averages 2 units; B averages 3 units per case.
3. Compare trade-off	B saves one second but costs one extra unit per case.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

Change: Require at most 2 cost units per case on average.

Prediction: A meets the budget; B does not.

Reason: The mandatory resource condition changes the decision.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Matched workload	The same task cases used for comparison
Median	The middle of ordered values under a declared convention
Tail latency	Slow outcomes hidden by typical-value summaries

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Performance and cost experiments

The facts we will use

AI ACADEMY / AI-6/29

Mean resource use per case

	Seconds/case	Cost units/case
A	3	2
B	2	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/29

Performance and cost experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute mean time

A averages 3 seconds; B averages 2 seconds per case.



2. Compute mean cost

A averages 2 units; B averages 3 units per case.



3. Compare trade-off

B saves one second but costs one extra unit per case.

CHANGE ONE THING

Require at most 2 cost units per case.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Performance and cost experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Resource improvement under one workload is not universal efficiency or equal quality.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?

A averages 3 seconds; B averages 2 seconds per case. A averages 2 units; B averages 3 units per case. B saves one second but costs one extra unit per case. Changed case: A meets the budget; B does not.

The mandatory resource condition changes the decision.

Changed-case reasoning: A meets the budget; B does not. The mandatory resource condition changes the decision.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main A mean 3/median 1/max 9 seconds; B mean/median/max 2; both 4/4 quality. Picture A averages 3 seconds/2 units, B 2 seconds/3 units; only A meets mean-cost budget ≤ 2. No individual cap is established. Selected C mean 4 and median 2.5; D mean/median 3 and quality 3/4. Retain the failed response's timing and quality flag; do not discard it or invent C's unprovided quality status.

Require scope and objective before preferring a configuration.

Diagnose the current misconceptions

The worked result establishes every future case: Resource improvement under one workload is not universal efficiency or equal quality.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

Teacher explanation: C mean 4, **median** 2.5; D mean 3, **median** 3. D quality 3/4, so speed comparison must include its failure.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Performance experiments compare **matched workload**s with declared timing boundaries and quality constraints. Averages can hide slow cases.

Use the same cases and environment; record every duration and success. For a simple **median** sort values and average the middle pair for even counts. Keep failed requests visible.

W2 Trace the worked case

Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

W3 Audit the limit

A report omits this qualification: Resource improvement under one workload is not universal efficiency or equal quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Resource improvement under one workload is not universal efficiency or equal quality.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Resource improvement under one workload is not universal efficiency or equal quality.

The new case should test this particular **mechanism** rather than merely restate the original result.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

Qualification: Resource improvement under one workload is not universal efficiency or equal quality.
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Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

Teacher explanation: C mean 4, **median** 2.5; D mean 3, **median** 3. D quality 3/4, so speed comparison must include its failure.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: C mean 4, **median** 2.5; D mean 3, **median** 3. D quality 3/4, so speed comparison must include its failure. Require intermediate reasoning and the stated scope. Fresh retry: E=[2, 2, 6, 10], F=[4, 4, 4, 4], all pass.

Expected reasoning: E mean 5, **median** 4; F mean 4, **median** 4. Tail and matched boundary remain relevant. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled performance and cost experiments **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of performance and cost experiments.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Security and privacy review. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/30, ai-6/32, ai-6/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of performance and cost experiments

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: A mean 3, **median** 1; B mean 2, **median** 2. B lowers mean but A has lower **median**; show the 9-second tail instead of declaring one universal winner.

Check every learner before discussion: Use the worked case for Performance and cost experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Resource improvement under one workload is not universal efficiency or equal quality.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Require at most 2 cost units per case on average. Require a **prediction** and a reason before comparing. Expected result: A meets the budget; B does not. Reason: The mandatory resource condition changes the decision.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Performance and cost experiments: the case



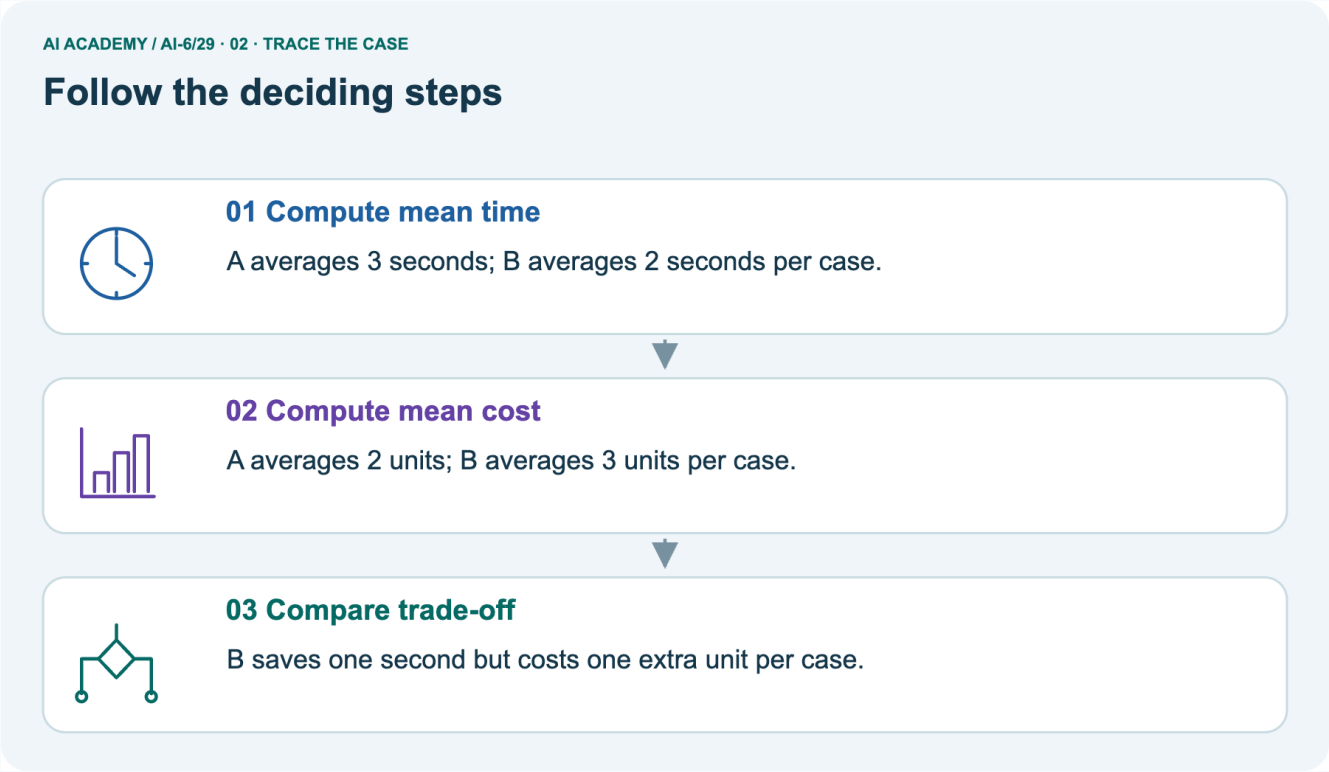
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric. Supplied facts: A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



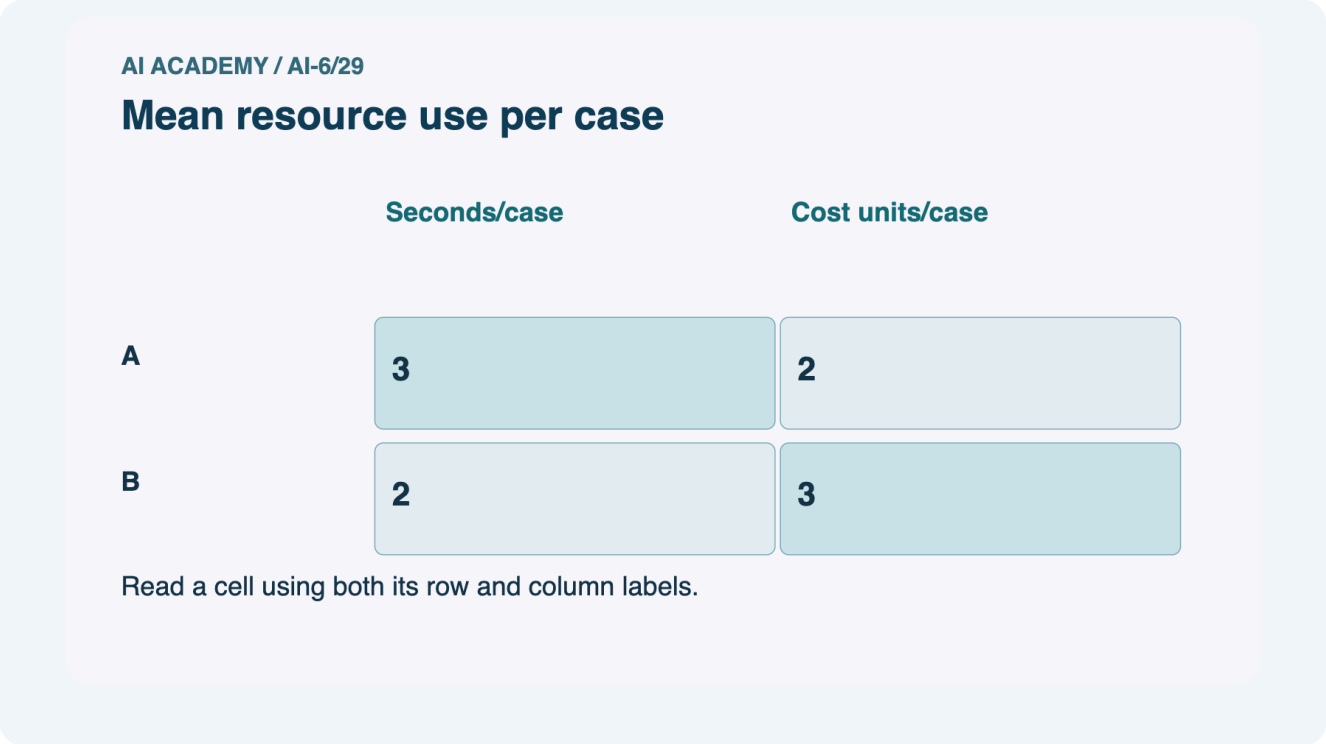
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute mean time: A averages 3 seconds; B averages 2 seconds per case. → Compute mean cost: A averages 2 units; B averages 3 units per case. → Compare trade-off: B saves one second but costs one extra unit per case.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Mean resource use per case



Follow the labelled picture with the matching explanation. Open the vector diagram

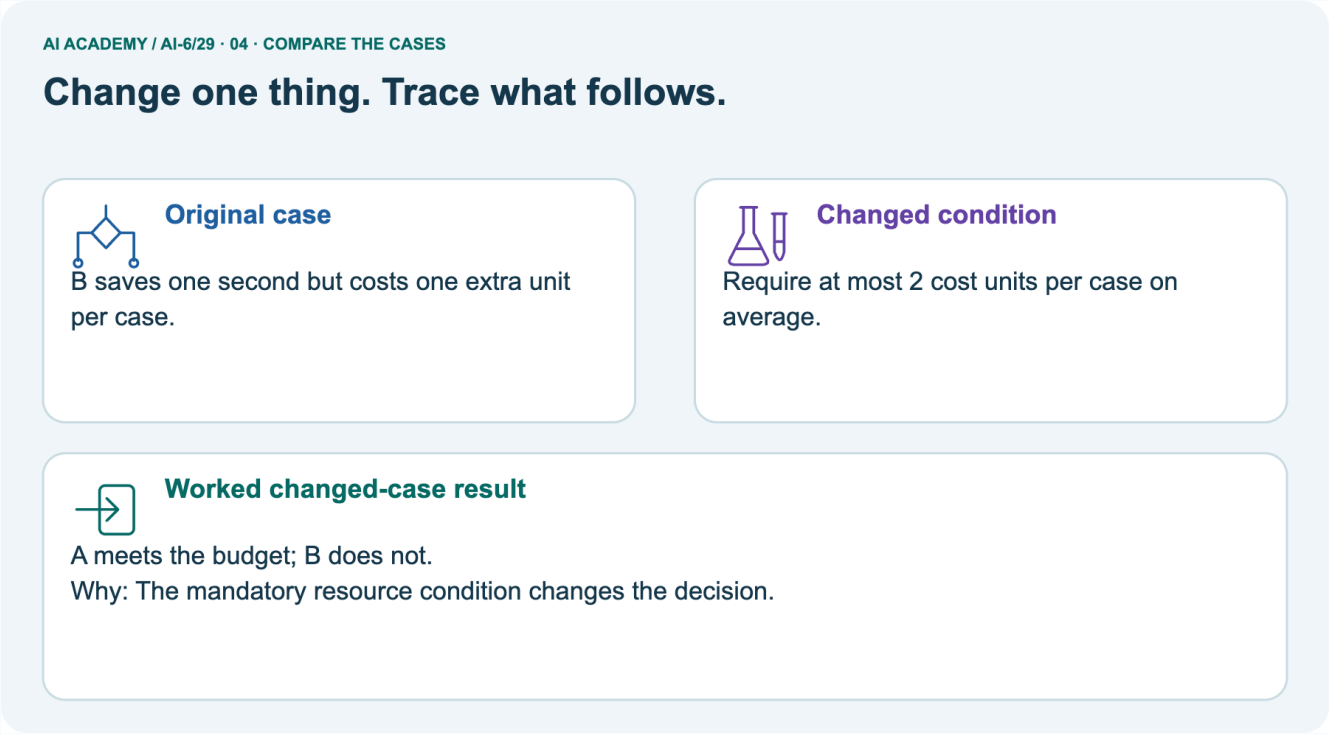
Mean resource use per case. A: Seconds/case = 3, Cost units/case = 2. B: Seconds/case = 2, Cost units/case = 3.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

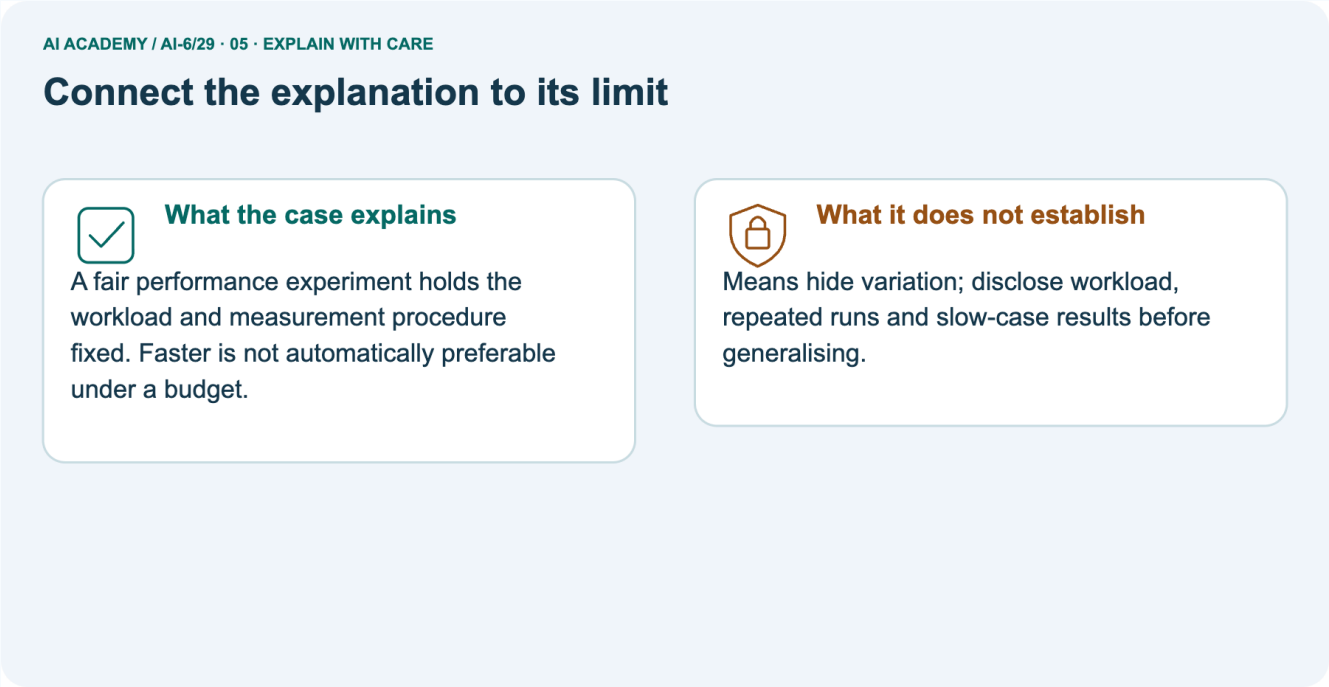
Original case: B saves one second but costs one extra unit per case. Changed condition: Require at most 2 cost units per case on average. Expected result: A meets the budget; B does not. Reason: The mandatory resource condition changes the decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The mandatory resource condition changes the decision.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

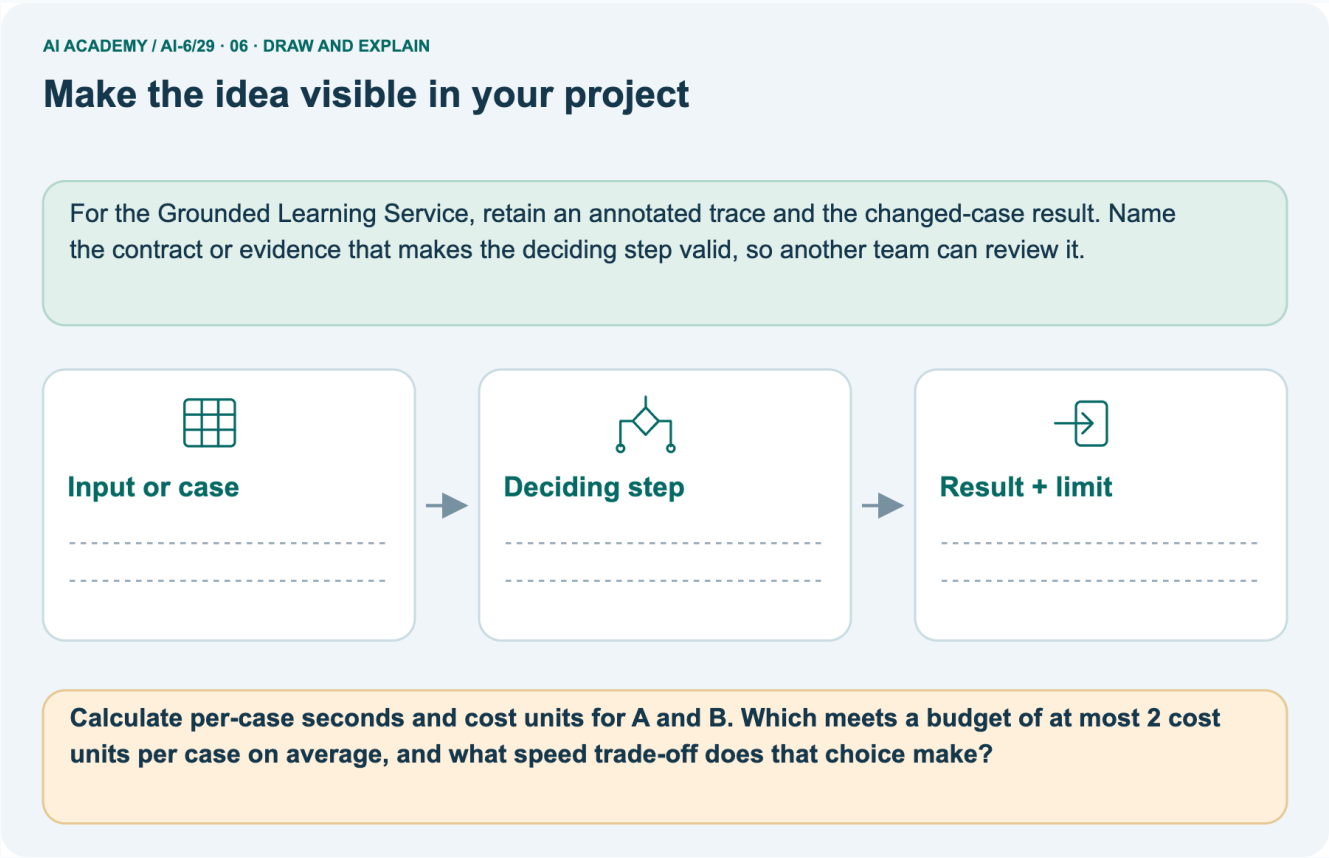
Limit: Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it. Draw your **input**, deciding step, result and limit. Supported question: Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A averages 3 seconds; B averages 2 seconds per case. A averages 2 units; B averages 3 units per case. B saves one second but costs one extra unit per case. Changed case: A meets the budget; B does not.

The mandatory resource condition changes the decision.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Performance and cost experiments: the case

02. Follow the deciding steps

03. Mean resource use per case

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.